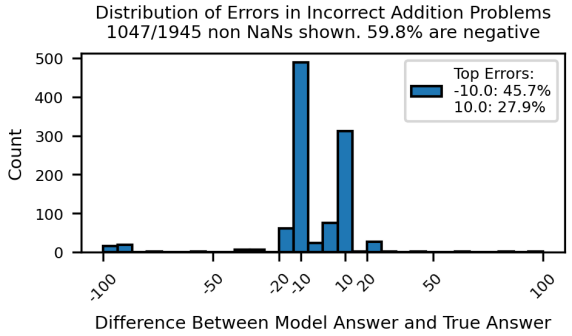


Figure 37. When GPT-J incorrectly answers an addition prompt (19.5% of the time), it answers with a number more than half the time. That number is usually off by 10 or -10 from the correct answer.

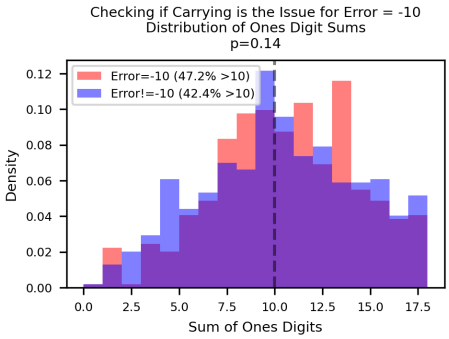


Figure 38. If GPT-J was struggling to "carry" when creating the $a + b$ helix, we would expect the ones digit of a and b to sum up to greater than 10 when the model is off by -10. However, we see that this is not significantly more likely for when the error is -10 than when it is not -10.

tigating at the neuron level. We sort neurons by their direct effect, and take the $k = 693$ highest DE neurons required to achieve 80% of the total direct effect (Fig 33). Then, we use the technique of LogitLens to understand how each neuron's contribution boosts and suppresses certain answers (see nostalgebraist (2020) for additional details). For the tokens $[0, 198]$ (the answer space to $a + b$), we see that each top DE neuron typically boosts and suppresses tokens periodically (Fig. 39). Moreover, when we Fourier decompose the LogitLens of the max activating $a + b$ example for each neuron, we find that a neuron whose preactivation fit's largest term is $c_{T_i,a+b}$ in Eq. 3 often has LogitLens with dominant period of T_i as well (Fig. 40). We interpret this as neurons boosting and suppressing tokens with a similar periodicity that they read from the residual stream helix with.

Despite being periodic, the neuron LogitLens are complex and not well modeled by simple trigonometric functions. Instead, we turn to more broadly looking at the model's final logits for each problem $a + b$ over the possible answer tokens $[0, 198]$. We note a similar distinct periodicity in Fig. 41. When we Fourier decompose the logits for all problems $a + b$, we find that the most common top period is 10 (Fig. 42). Thus, it is sensible that the most common error is ± 10, since $a + b - 10$, $a + b + 10$ are also strongly promoted by the model. To explain why -10 is a more common error than 10, we fit a line of best fit through the model logits for all $a + b$, and note that the best fit line almost always has negative slope (Fig. 42), indicating a preference for smaller answers. This bias towards smaller answers explains why GPT-J usually makes mistakes with larger a and b values (Fig. 11, Appendix A).

G. Tooling and Compute

We used the Python library `nnsight` to perform intervention experiments on language models (Fiotto-Kaufman et al., 2024). All experiments were run on a single NVIDIA RTX A6000 GPU with 48GB of VRAM. With this configuration, all experiments can be reproduced in two days.

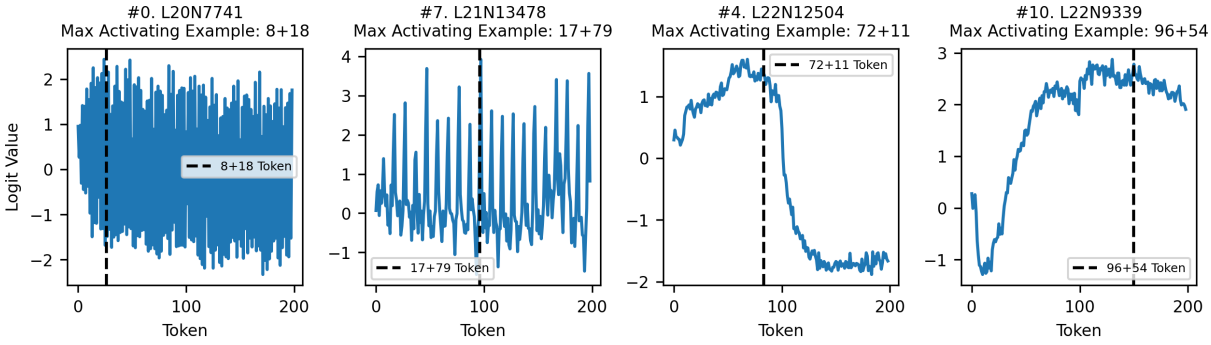


Figure 39. Using the LogitLens technique, we analyze the contributions of the top neurons presented in Fig. 8 for each neuron's maximally activating $a + b$ example.

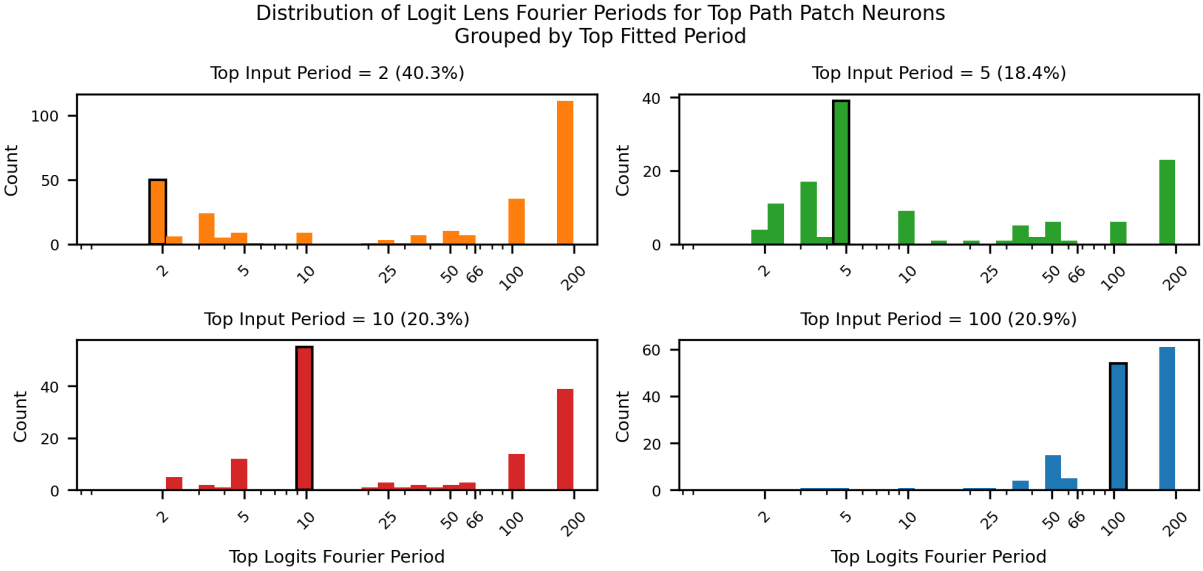


Figure 40. For all neurons with top $a + b$ fit component with $T = [2, 5, 10, 100]$, we plot the distribution of the top Fourier period in their LogitLens taken over the tokens $[0, 198]$. We see that a neuron with top fitted period of T_i often has a LogitLens with top Fourier period T_i. Surprisingly, 200 is a common Fourier period, possibly used to differentiate numbers in $[0, 99]$ from $[100, 198]$.

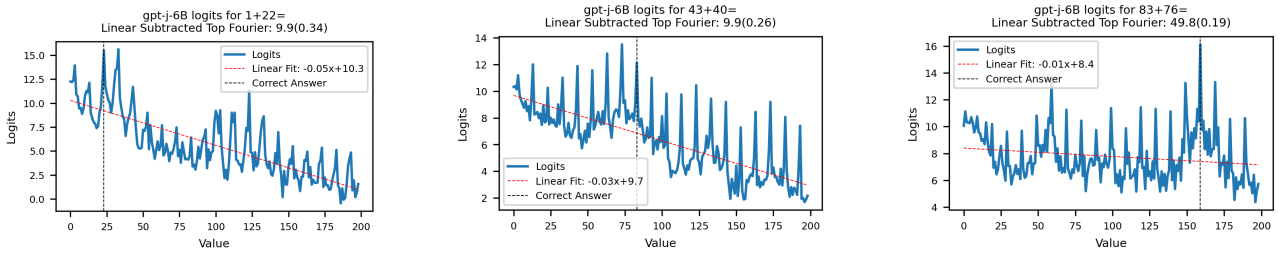


Figure 41. We plot the final model logits over the token space $[0, 198]$ for some randomly selected examples. We see clear periodicity with a sharp period of 10, in addition to a general downward trend indicating a preference for smaller answers.

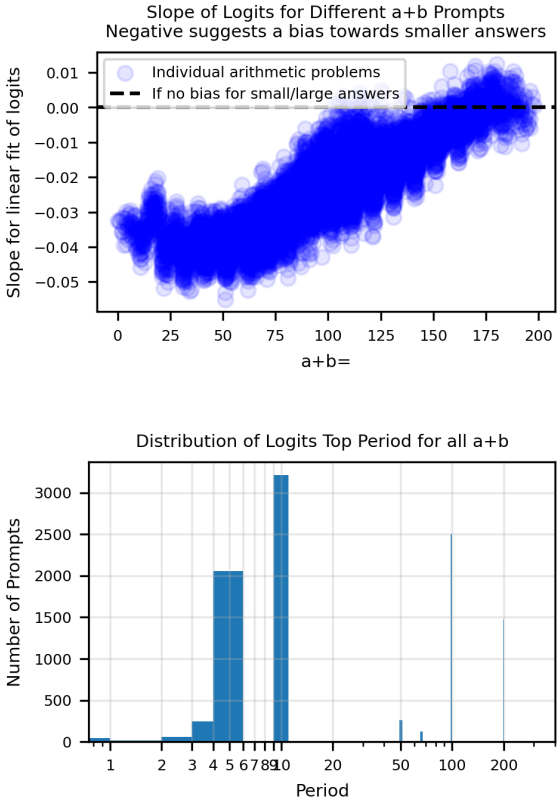


Figure 42. Top When we plot the slope of the best fit line over all logits, we see that the slope is often negative, implying a bias towards smaller answers. *Bottom* When applying a Fourier decomposition on the logits for all examples $a + b$ over the token space $[0, 198]$, we see that 10 is the most common period. Note that we subtract out the fitted linear component first before applying the Fourier transform.